

RESEARCH INTERESTS	I study how algorithms and incentives can be designed and tuned automatically. One thread of my work is algorithm configuration and selection: using empirical performance models and learning to make solvers for hard combinatorial and Boolean reasoning problems faster. The other is algorithmic game theory, where computation meets strategic behavior in mechanism design and multi-agent settings. I aim to combine formal reasoning with learning to build systems that are efficient and interpretable.	
EDUCATION	University of British Columbia	Vancouver, Canada
	<i>Ph.D. in Computer Science</i>	
	<i>Supervisor: Prof. Kevin Leyton-Brown</i>	Sep 2026 - Present
	Research areas: Algorithm Configuration, AI for Combinatorial Optimization	
	Sabancı University	Istanbul, Turkey
	<i>B.S. in Computer Science and Engineering, B.S. in Industrial Engineering</i>	
	<i>Minor in Mathematics</i>	Sep 2021 - June 2026
	- GPA: 3.98/4.00, Top 3 GPA in the Faculty of Engineering and Natural Sciences - Full Tuition Scholarship	
	Uppsala University (Erasmus Exchange)	Uppsala, Sweden
	<i>Computer Science and Mathematics</i>	Jan 2024 - June 2024
RESEARCH EXPERIENCE	CON-NET: Platform-Agnostic Bot Detection Model	
	<i>TUBITAK & CHIST-ERA funded project</i>	
	<i>Supervisor: Dr. Onur Varol</i>	Jan 2025 - Feb 2026
	- Designing and implementing a platform-agnostic bot detection model - Implementing account-based and digital-dna based VAE models - Feature engineering for cross-platform compatibility and implementation of novel approaches (e.g. L2P and SpringRank) to avoid distribution shifts	
	Partial Observable Multi-Agent Path Finding with Constrained Communication	
	<i>Erasmus Internship, University of Luxembourg @ ICR Group</i>	
	<i>Supervisors: Prof. Leon van der Torre & Dr. Pere Pardo</i>	July - Sep 2025
	- Designed a framework to solve a novel variant of the Multi-Agent Path Finding (MAPF) problem featuring partial observability, decentralization, and anonymous task-sequential requirements with constrained communication - Modeled observation and communication using dynamic epistemic logic and employed Answer Set Programming for collaborative planning under uncertainty - Developed a simulation script to compare performance across different exploration strategies and map configurations	
	Solving Matching Problems Using AI Methods	
	<i>PURE Project, Sabancı University @ CogRobo Lab</i>	
	<i>Supervisor: Prof. Esra Erdem</i>	Sep 2024 - Jan 2025
	Designed a student-to-department assignment system using Gale-Shapley algorithm and Constraint Programming to balance student preferences, diversity goals, and department capacities	
	Automatic Transcription of Ottoman Documents (AKIS)	
	<i>TUBITAK funded project, Sabancı University @ DH Lab & VPA Lab</i>	
	<i>Supervisors: Prof. Berrin Yanıkoglu & Dr. Esmâ Bilgin Tasdemir</i>	Aug 2023 - Jan 2024
	Enhanced image segmentation capabilities of existing R-CNN model and achieved significant improvement in model accuracy for document transcription	
	A Digital Analysis of an Early Ottoman Chronicle	
	<i>PURE Project, Sabancı University @ DH Lab</i>	
	<i>Supervisors: Dr. İnanç Arın & Dr. Mehmet Kuru</i>	July - Sep 2023

	- Created geo-visualizations and interactive data analyses of Asikpasazade's Chronicle - Performed NLP analyses with BERT and BERTopic to explore sentiment and themes - Developed an LLM-based chatbot using GPT-3.5 to enable dynamic user interaction with the chronicle
INDUSTRY EXPERIENCE	Legotize: 3D to LEGO Converter <i>Valensas Software, Supervisor: Akın Idil</i> Aug - Sep 2024 - Designed a program to convert 3D models into pixelated LEGO representations using PyTorch3D. Vinventory: Inventory Management App <i>Valensas Software, Supervisor: Moray Baruh</i> July - Sep 2024 - Built an inventory management app using Go, React.js, and PostgreSQL with Azure AD authentication. Integrated CI/CD pipelines with GoReleaser and deployed on Kubernetes with Helm Charts. - The app is currently being used as a tool inside the company. OREDATA Data Science Intern Istanbul, Turkey Jan 2023 - Feb 2023 - Experimented with and compared different tumor-detection ML and DL models using scikit-learn. Practiced exploratory data analysis leveraging Kaggle datasets.
TEACHING EXPERIENCE	Teaching Assistant: Introduction to Artificial Intelligence (AI 100, University of British Columbia, Sep 2026 - present); Theory of Computation (Prof. Kemal Inan, Jan 2026 - June 2026); Algorithms (Prof. Esra Erdem, Jan 2025 - Jan 2026); Machine Learning (Dr. Onur Varol, Sep 2024 - Jan 2025); Data Structures (Dr. Gülsen Demiröz, Sep 2023 - Jan 2024).
AWARDS AND HONORS	Ranked 139th among 2,4+ million students in Turkey's national university entrance exam 2021 Full Tuition Scholarship and Monthly Stipend, Sabanci University 2021 TUBITAK 1001 Undergraduate Scholarship, Research Trainee on a TUBITAK-funded project with monthly stipend Aug 2023 – Jan 2024 Top 3 GPA, Faculty of Engineering and Natural Sciences, Sabanci University 2026
SKILLS	Languages: Turkish (Native), English (IELTS: 8.0/9.0), German (Basic) Programming: Python, C/C++, Go, JavaScript, Java, Verilog ML/AI: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, NetworkX, Matplotlib, Seaborn, PyTorch3D Tools & Infrastructure: Git, Docker, Kubernetes, Flask, MERN Stack, Unix/Linux CLI, L^AT_EX, Markdown Databases: PostgreSQL, MySQL, NoSQL
EXTRA-CURRICULAR ACTIVITIES	Google Developer Student Club: Led workshops on TensorFlow, Selenium and Git/GitHub for 2-3 hours each. Literature Club: Board member of the Literature Club of Sabancı University for 4 years, the last year as the President. For the last 3 years, I was the coeditor of the club's semiannual literature fanzine, "harmoni." Social Responsibility: Volunteered in shoreline cleanups, Gender and Memory Walks, and botanic garden projects.